CIS4020 - Course Project
Report Due: 4 December 2026, 5pm EST

Your team takes a real-world dataset from **raw data to a working, demonstrable solution**, then communicates it professionally. The project spans the course, working through the full data science pipeline (framing, data wrangling, EDA, modelling, honest evaluation, and communication) and ends in a **live demo** in class.

You choose the **domain** (and dataset). Pick from the curated menu below, or **propose your own with instructor approval**. Whatever you choose, every team completes the same **required pipeline** and the same three **deliverables**, so grading stays fair.

Learning objectives

- **Frame** an open-ended problem as a well-posed data science question with a success metric.
- **Prepare** a real, messy dataset, justifying your cleaning, missing-data, and outlier decisions.
- **Explore** the data to extract evidence-backed findings through clear visualizations.
- **Model** the problem from a baseline up to a justified stronger approach.
- **Evaluate** models with task-appropriate metrics and a sound validation scheme.
- **Interpret** results, addressing limitations, and ethical implications.
- **Communicate** the work in a written report and a live demo.
- **Collaborate** as a team, integrating individual contributions into a coherent whole.

Required pipeline (every team, regardless of domain)

This shared spine is what makes projects comparable and gradeable.

1. **Problem framing**: a clear question, why it matters, and a success metric chosen *before* modelling.
2. **Data acquisition & understanding**: obtain the data; document its provenance, structure, size, and limitations; classify key attributes by type.
3. **Cleaning & preparation**: missing values, outliers, inconsistencies, joins/merges, with *justified* decisions.

4. **Exploratory data analysis**: insight-driven, with clear, honest visualizations.
5. **Feature engineering**: building useful inputs from the raw columns.
6. **Modelling**: a trivial **baseline** plus at least one stronger model, with a justified choice.
7. **Evaluation**: appropriate metrics, a sound validation scheme, and an **error analysis**.
8. **A working demo**: an interactive demonstration of the result.
9. **Reflection**: limitations, ethics, and what you'd do with more time.

Datasets

Pick a dataset from the following domains or propose your own based on the requirements below.

Healthcare

- **Diabetes 130-US Hospitals** UCI #296 The dataset represents ten years (1999-2008) of clinical care at 130 US hospitals and integrated delivery networks. Each row concerns hospital records of patients diagnosed with diabetes, who underwent laboratory, medications, and stayed up to 14 days.
- **CDC Diabetes Health Indicators** UCI #891 The Diabetes Health Indicators Dataset contains healthcare statistics and lifestyle survey information about people in general along with their diagnosis of diabetes. The 35 features consist of some demographics, lab test results, and answers to survey questions for each patient.

Agriculture

- **Crop Yield with Soil & Weather (India, 1997–2020)** This dataset collection combines crop yields, soil composition, and weather data across Indian states, providing a comprehensive view of agriculture between 1997 and 2020.

Sports analytics

- **Lahman Baseball Database** The Lahman Baseball Database, created by SABR member Sean Lahman, contains complete major league batting and pitching statistics back to 1871
- **StatsBomb Open Data** StatsBomb are committed to sharing new data and research publicly to enhance understanding of the game of Football.

Financial

- **Default of Credit Card Clients** UCI #350 This dataset represents a collection of customers' default payments in Taiwan.

Education

- **Predict Students' Dropout & Academic Success** UCI #697 The dataset was created in a project that aims to contribute to the reduction of academic dropout and failure in higher education, by using machine learning techniques to identify students at risk at an early stage of their academic path, so that strategies to support them can be put into place.

Biology / genetics

- **Gene Expression Cancer RNA-Seq** UCI #401 This collection of data is part of the RNA-Seq (HiSeq) PANCAN data set, it is a random extraction of gene expressions of patients having different types of tumor: BRCA, KIRC, COAD, LUAD and PRAD.
- **Mice Protein Expression** UCI #342 Expression levels of 77 proteins measured in the cerebral cortex of 8 classes of control and Down syndrome mice exposed to context fear conditioning, a task used to assess associative learning.

Propose your own (with approval)

You may bring a different dataset (e.g., from the awesome-public-datasets repository) if it meets **all** of these:

- **Moderate size**: roughly 1,000–500,000 rows and under a few hundred MB (runs in pandas on a laptop). Fewer rows is fine only if the data is **rich in attributes**;
- **Enough variety**: at least 10 meaningful columns/attributes with a diversity of types.
- **A clear question**: an attribute you could predict, or expected structure to find.
- **Real and messy**: needs actual cleaning (missing values, mixed types, inconsistencies, or joins). Pre-cleaned toy sets (e.g., Iris) do not qualify.
- **Documented & accessible**: free; a working download link to a concrete file (CSV / JSON / SQLite / Parquet); a data dictionary; no login/credentialing wall; license allows academic use.

Deliverables

1. Project Proposal (1–2 pages) — due Oct 23

Submit a brief proposal that shows your project is well-scoped and feasible. State the **task** you plan to work on and **how you'll measure success**, the specific question you're answering and the metric or criterion you'll judge your results against. Describe the **dataset**: its source, approximate size (rows × columns), the key fields you'll use, and how you'll obtain it. Include a short **loaded sample** (e.g., the output of df.head() and df.shape) to show you actually have the data in hand. Give a **high-level planned approach** (the main steps from raw data to result). Identify your **team and each member's role**, and lay out a brief **timeline** with milestones from now to the final

deliverables. Keep it concise and concrete; this is a plan we can give feedback on, not a final commitment.

Your proposal must include:

- **Task & success measure**: the question you're tackling and how you'll know if you've succeeded.
- **Dataset**: source, size, key fields, how you'll obtain it, and a loaded sample.
- **Planned approach**: a high-level outline of your plan.
- **Team & roles**: who's on the team and who's responsible for what.
- **Timeline**: milestones through to the final submission.

2. Presentation & demo — Week 12-13

In class presentation:

- 10–15 minutes
- Problem, data, key EDA findings, modelling choices, results, and limitations
- A live demo (i.e., your Jupyter notebook)
- All members present

3. Final report + repository — Due Dec 4

A report (8–12 pages) + a **reproducible notebook and repository**: every pipeline stage documented with justified decisions; results including what didn't work; ethics & limitations; a clean README + environment file; and a **personal contribution statement**.

Teamwork & academic integrity

- A **contribution statement** in the report is required.
- You may use generative-AI tools, but you must **disclose how**, and understand and be able to explain every line you submit. Don't submit results you can't reproduce from your own code. Copying another team's work is not permitted.

Grading

The project counts for 40% of your course grade. The share of the project for each component is given in the table below:

Component	Share of project
Proposal	12.5%
Presentation & demo	25%
Final report + repo	62.5%

Project report rubric (100 points):

- Problem statement & success metric (10)
- Data wrangling (25)
- EDA (10)
- Modelling + baseline + justification (20)
- Evaluation + error analysis (20)
- Ethics & limitations (10)
- Reproducibility (5)